WARNING As you learned in chapter 4 when validating anti-CSRF tokens, it is important to always use a constant-time equality when comparing a secret value (the correct authentication tag) against a user-supplied value. Timing attacks against HMAC tag validation are a common vulnerability, so it is critical that you use `MessageDigest.isEqual` or an equivalent constant-time equality function.

Listing 5.11 Validating the HMAC tag

```
@Override
public Optional<Token> read(Request request, String tokenId)
{
    var index = tokenId.lastIndexOf('.');    #A
    if (index == -1) {     #A
        return Optional.empty();    #A
    }    #A
    var realTokenId = tokenId.substring(0, index);    #A

    var provided = Base64url.decode(tokenId.substring(index +
1));    #B
    var computed = hmac(realTokenId);    #B

    if (!MessageDigest.isEqual(provided, computed)) {    #C
        return Optional.empty();
    }

    return delegate.read(request, realTokenId);    #D
}
```

#A Extract the tag from the end of the token ID. If not found, then reject the request.
#B Decode the tag from the token and compute the correct tag.